

Order task: one window over frames 16 to 147, so the baseline and the trained baselines have 48 to 54 features and are read as they are; the reservoirs' 3,072 to 6,144 are projected to 192.



mel spectrogram:
16 band energies
per frame

Spectrogram-only baseline

no reservoir:
the 16 band energies
themselves

mean, standard deviation and
mean absolute change of each
signal over 4 windows:
192 features

standardize
(already 192 wide)

Reservoir, untrained: an oscillator network or a leaky-integrator bank

e.g. 1,024 oscillators,
sine and cosine of each
phase: 2,048 signals

mean, standard deviation and
mean absolute change of each
signal over 4 windows:
12,288 to 24,576 features

standardize, then a
random projection to 192

Trained baseline: GRU, TCN, CNN, transformer or S4D

a small network trained
end to end: 16 hidden
units (GRU: 18)

mean, standard deviation and
mean absolute change of each
signal over 4 windows:
192 features
(GRU: 216)

standardize
(GRU: projected to 192)

ridge readout:
one linear layer,
192 features
→ 10 digit scores,
fitted in closed
form, the same
for every arm

digit

gradients train
the network

learned linear head
(training only)